

COMP1017 (Mathematics for Computer Scientists) Test 3

Academic Session 2017-2018, Autumn Semester

There are TWO (2) questions pertaining to numbers, matrices, series, and induction on ONE (1) page. Please answer ALL the questions. Time allowed 30 minutes.

1. Let S be the set of first N positive integers. Let $a, b \in S$, and $m, n < N$ be positive integers.

(a) If $m|a$ and $m|(a - b)$, is it true that $m|b$? Under what condition is m the greatest common divisor of a and $a - b$? Show your work.

[16 marks]

(b) Let two positive integers $m, n < N$ be relatively prime, i.e., $\gcd(m, n) = 1$. Assume $a \neq b$. Define two pairs as $(i_a, j_a) = (a \bmod m, a \bmod n)$ and $(i_b, j_b) = (b \bmod m, b \bmod n)$, respectively. Under what circumstances will (i_a, j_a) and (i_b, j_b) uniquely represent a and b respectively (i.e., $(i_a, j_a) \neq (i_b, j_b)$)? Explain.

[16 marks]

(c) Let $N = 30, m = 7$. Let A be a matrix. Design a method to assign each positive integer in S to exactly one entry in A , such that all the integers congruent modulo 7 are in the same row, whilst the integers not congruent modulo 7 are in different rows. Specify the minimum size of A . Show your work.

[18 marks]

2. Let i, j, n be integers in the following questions.

(a) Calculate the summations $\sum_{i=3}^7 5$, $\sum_{i=1}^2 \sum_{j=i}^4 (i - j)$, and $\sum_{x \text{ is prime } \wedge x < 7} x^2$, respectively. Show your work.

[16 marks]

(b) Using the mathematical induction, prove that for all $n \geq 7$, $11n + 17 \leq 2^n$.

[18 marks]

(c) If n people stand in a line, where n is positive, and if the first person in the line is a woman and the last person in line is a man, then somewhere in the line there is a woman directly in front of a man. Specify the basis step and the inductive step respectively if a mathematical induction is used to prove the statement. (You do not need to prove the statement.)

[16 marks]